

Ideally symmetrical systems.

The oscillatory model of Universe.

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Abstract

This article presents a speculative relational interpretation of symmetry, arguing that dynamics arise from asymmetry, while states of ideal symmetry are inherently unstable. Symmetry is treated as a property of relations between system elements rather than of objects or fields, and even minimal relational disturbances lead to spontaneous symmetry breaking. The proposed Theory of Relational Symmetry Instability (TRSI), formulated in a speculative and interpretative framework, is consistent with Noether's theorem, quantum field theory, and early-Universe cosmology. Applied to cosmology, the article suggests a self-propelling, cyclic evolution of the Universe driven by the tension between the tendency toward symmetry and the intrinsic instability of perfectly symmetric states.

Keywords: symmetry, asymmetry, system dynamics, relational symmetry, instability, Universe

1. Introduction

During the development of a model of time perception, difficulties were encountered related to the unambiguous definition of the very concept of time. In the article "Time as an Emergent Phenomenon. Symmetry and Asymmetry of Systems" (Leiss, 2025), a theory was proposed according to which time constitutes an emergent phenomenon resulting from changes occurring in dynamic systems, with dynamics defined in a relational rather than temporal manner. As the primary cause of change, asymmetry of the system was indicated, which led to the formulation of the thesis that asymmetric systems are inherently dynamic, whereas ideally symmetrical systems cannot occur in physical reality. The present article develops this concept, focusing on an analysis of the nature of symmetrical systems and the mechanisms leading to their instability.

The proposed theory of relational symmetry instability constitutes an approach consistent with current knowledge while remaining original in its treatment of the phenomenon of symmetry breaking. In contrast to standard interpretations that attribute symmetry to objects or fields, this theory treats it as a property of the relations between elements of a system. This leads to the conclusion that **ideally symmetrical systems are inherently unstable, because they require absolute homogeneity of relations, which cannot be maintained under real physical conditions.** A minimal relational disturbance is sufficient to cause spontaneous symmetry breaking. This approach remains consistent with field theory, Noether's theorem, and contemporary cosmology of the early Universe, in particular with the mechanisms of generating inflationary fluctuations and the dynamics of the quantum vacuum. As a result, the theory of relational symmetry instability provides a useful interpretative tool that allows the genesis of asymmetry and structures in the Universe to be described in a new way.

The present work does not aim to formulate new physical theories, but constitutes a summary of available knowledge and is **speculative** in nature; the equations used serve solely an illustrative function, enabling a clearer presentation of the essence of the issue discussed. Its main goal is a reinterpretation of known mechanisms in a **relational framework**.

2. Symmetry

When analyzing the concept of symmetry, one may use the analogy of a balance scale on which two identical objects are placed and subjected to the same interactions. In an intuitive view, the state of ideal symmetry signifies the absence of a dynamic process, because both elements remain in complete equilibrium. However, a more detailed analysis indicates that the least stable element of such a system is not the objects themselves, but the relational structure that connects them, in this case the scale. Under conditions of ideal equilibrium, deprived of the possibility of minimal deviation, the relation connecting the structures becomes susceptible to instability and consequently undergoes destabilization. This metaphor reflects the main thesis of the present theory: it is not the objects forming a symmetrical system that determine its stability, but the relations between them, which, in a state of ideal symmetry, lead to spontaneous symmetry breaking.

2.1. Theoretical foundations

Symmetry plays a fundamental role in classical and quantum physics as well as in the cosmology of the early Universe. The literature emphasizes the importance of both the symmetry of objects and their mutual relations in shaping the dynamics of physical systems (Guth, 1981; Coleman & Weinberg, 1973; Linde, 1982). Existing inflationary models assume an initial homogeneity and isotropy of the Universe, corresponding to ideal symmetry in terms of energy density and field properties (Albrecht & Steinhardt, 1982; Piazza & Vernizzi, 2013). The mechanism of spontaneous symmetry breaking, as described in the classic works of Coleman and Weinberg (1973), indicates that even in formally symmetrical systems, quantum fluctuations may lead to the selection of one of many equivalent configurations, resulting in the emergence of observable structures. Research on the process of “reheating” after inflation (Kofman, Linde & Starobinsky, 1997) and the role of topological defects in eternal inflation (Linde & Linde, 1994) further shows that the symmetry of the early Universe may undergo dynamic breaking, leading to the formation of rich cosmic structure. Analyses of the global structure of inflationary models (Linde & Mezhlumian, 1993) and the singularity theorems of Penrose and Hawking (1970) indicate that even with the preservation of formal symmetry of physical laws, the realization of a system in a specific state is unstable.

Recent works deepen this perspective. Khan, Kim, and Ko (2024) demonstrated that combining inflation with the breaking of dark $U(1)$ symmetry can simultaneously generate Higgs particle masses and account for dark matter. Sagunski, Schicho, and Schmitt (2023) showed that the breaking of conformal symmetry in QCD in the early Universe can lead to the generation of observable gravitational waves, while Bettoni and Rubio (2021) described inflationary models in which scale symmetry appears and is broken in different cosmological epochs. In the context of baryogenesis, Pereira, Ferraz, Lobo, and Mimoso (2024) emphasize that the breaking of C and CP symmetries and their interactions in extended gravitational theories constitute a key mechanism for the formation of matter. All these results support the hypothesis that symmetry breaking in early-Universe cosmology is a relational process, dependent on mutual configurations of fields and particles, rather than solely on the asymmetry of objects themselves.

Julian Barbour develops a relational approach to physics in which time and space are not absolute entities, but emerge from configurations of relations between elements of a system. In his interpretation, dynamics consists of changes of relations, not evolution in an independent time (Barbour, 1999). This approach is philosophically close to interpretations of relational symmetry, although it differs in emphasis: whereas Barbour focuses on eliminating absolute notions, the approach discussed here emphasizes the internal instability of ideally symmetrical states and their inevitable disturbance.

2.2. Symmetry in a relational perspective

According to our theory, the primary source of instability in ideally symmetrical systems is **not the objects themselves, but the relations between them**. Even when the elements of a system remain identical and preserve their symmetry, minimal disturbances in their relation lead to a change in the system’s configuration, breaking relational symmetry. In a cosmological model, this means that the initially homogeneous and isotropic vacuum of the Universe could have

undergone spontaneous symmetry breaking not due to asymmetry of physical laws, but as a result of **instability in the relations between elements**. Such an interpretation provides a coherent explanation for both the emergence of cosmic structures and fluctuations.

3. Noether's Theorem

In light of Noether's theorem, according to which each continuous symmetry of the action corresponds to a specific conservation law, the proposed interpretation of symmetry as a relational property allows for a new approach to the instability of ideally symmetrical systems. In such a system, the degeneracy of many states—for example with the same energy—means that stability depends not on the properties of the objects themselves, but on the homogeneity of their mutual relations. The source of observed asymmetries is therefore not a violation of the symmetry of physical laws, but the dynamic instability of the relational structure.

In this perspective, the formation of structures in the early Universe becomes a natural consequence of the relational sensitivity of symmetrical systems, rather than their material or geometric imperfection. Symmetry breaking is an inherent mechanism by which symmetrical systems in classical and quantum physics transition from abstract formal symmetry to concrete, observable configurations, while preserving symmetry in physical laws but not in the realization of system states.

4. Theory of Relational Symmetry Instability (TRSI)

In the previous work, the thesis was formulated that ideally symmetrical systems do not occur in nature; however, analysis of cosmological history suggests that states of ideal symmetry could theoretically have appeared both at the dawn of the Universe and in its hypothetical final state. In this view, a state of primordial symmetry could constitute a necessary condition for the occurrence of the Big Bang, while an analogous symmetrical state in the distant future could be responsible for a potential cosmological collapse. It follows from this, in accordance with earlier argumentation (Leiss, 2025), that ideally symmetrical systems are internally unstable, and their breakdown leads to the emergence of dynamics. After undergoing a process of symmetry loss, as in the case of cosmic expansion initiated by the Big Bang, the cosmological system tends to regain symmetry. This tendency, in view of the inherent instability of the ideally symmetrical state, generates cyclic dynamics in which the Universe functions as a self-propelling system, alternately moving away from symmetry and returning toward it.

In summary, it can be observed that the dynamics of the Universe result from a fundamental property of physical systems, namely asymmetry. It enables processes to occur, generates a direction of evolution, and constitutes a condition for the existence of time as an emergent quantity. At the same time, the natural tendency of the cosmological system is to strive toward states with the highest possible degree of symmetry, which, according to the presented theory, leads to approaching internally unstable configurations. Consequently, the Universe never reaches a state of ideal equilibrium, but constantly oscillates between phases of approaching symmetry and losing it. This imbalance between the tendency toward symmetry and its spontaneous breakdown creates a mechanism that sustains the continuity of physical processes and ensures lasting dynamism of the cosmological system, without the need to invoke an external source of energy or cause.

Being the designer of a closed system that is to remain in continuous motion, that is, to be characterized by lasting dynamics, one can construct it based on three fundamental principles: first, asymmetry is the primary driver of dynamics, initiating changes in the system; second, ideally symmetrical systems are internally unstable, and any system that reaches full symmetry is immediately destabilized; third, the system strives toward a state of symmetry, the achievement of which automatically triggers instability and a return to a dynamic state. In this way, the system becomes self-propelling, and the continuity of processes and uninterrupted dynamics are a consequence of the inherent contradiction between the striving for symmetry and its internal instability.

5. Fundamental assumption:

The symmetry of a physical system is not a property of the objects themselves or their individual parameters, but a relation between the elements of the system (e.g., particles, fields, degrees of freedom).

Let us assume a system composed of two elements E_i and E_j .

Their symmetry depends on the relation $R(E_i, E_j)$, that is, a measure of dependence or mutual connection between the elements of the system; in a state of ideal symmetry this relation is completely homogeneous and shows no differentiation. If the system consists of elements E_i , then symmetry concerns the set of relations $R(E_i, E_j)$, not the elements themselves.

Symmetry (ideal state):

$$\|R(E_i, E_j)\| = 0$$

the relation between elements is perfectly balanced (full symmetry).

Instability of symmetry:

An ideally symmetrical system is physically unstable, because it requires the preservation of perfect homogeneity of relations between all elements.

Ideal symmetry is unstable, therefore even the smallest disturbance:

$$\|\Delta R(E_i, E_j)\| \neq 0$$

causes immediate symmetry breaking.

Asymmetry generates dynamics:

The dynamics of the system D (Leiss, 2005) is proportional to the degree of asymmetry:

$$D \propto \|R(E_i, E_j)\|$$

the greater the disturbance of relations, the greater the dynamics.

The system always strives to achieve ideal symmetry:

$$\|\Delta R(E_i, E_j)\| \rightarrow 0$$

However, when it reaches ideal symmetry:

$$\|R(E_i, E_j)\| = 0 \implies \|\Delta R(E_i, E_j)\| \neq 0$$

Symmetry itself causes an immediate deviation, that is, the beginning of a new asymmetry.

Ideal symmetry creates its own disturbance.

6. Conclusion

Symmetries in physical systems are structures of relations between elements, and their ideal realization is dynamically unstable. Spontaneous symmetry breaking results from relational instability, not from asymmetry of objects or physical laws.

7. Summary, further research directions, and limitations

The analysis of ideally symmetrical systems indicates that stability does not result from the properties of the elements themselves, but from the relations between them. Ideal symmetry is internally unstable; even minimal disturbances of relations lead to symmetry breaking and the generation of dynamics. In the context of cosmology, the theory of relational symmetry instability allows for a coherent interpretation of the emergence of fluctuations and structures in the early Universe, independently of asymmetry in physical laws, and enables a reinterpretation of classical mechanisms such as the Higgs field potential or inflationary dynamics.

One of the main advantages of this work is the possibility of explaining the continuous, dynamic character of the Universe in a closed-system framework. Assuming that the Universe is a self-sufficient system, TRSI makes it possible to understand its constant evolution as a natural effect of the inherent instability of ideally symmetrical states and the cyclic striving toward symmetry. In this approach, cosmic dynamics does not require an external source of energy; the continuity of processes and the emergence of structures are consequences of the fundamental relational properties of physical systems. This constitutes a new perspective on the evolution of the cosmos, in which symmetry breaking generates a self-propelling mechanism of motion and structuring of matter.

Further research should include quantum modeling of relational instability in multi-element systems, application of TRSI in simulations of the early Universe, investigation of the influence of relational instability on the evolution of topological defects, and development of the formalism in the context of open and thermodynamic systems.

The main limitations of the work include the simplification of the model to two-element systems and the illustration of relations in a theoretical manner. Real cosmological and quantum systems exhibit much greater complexity and many-body interactions. Moreover, the mathematical formalization requires further grounding within exact field theories and quantum mechanics in order to enable precise predictions of observable fluctuations and structures.

8. References

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